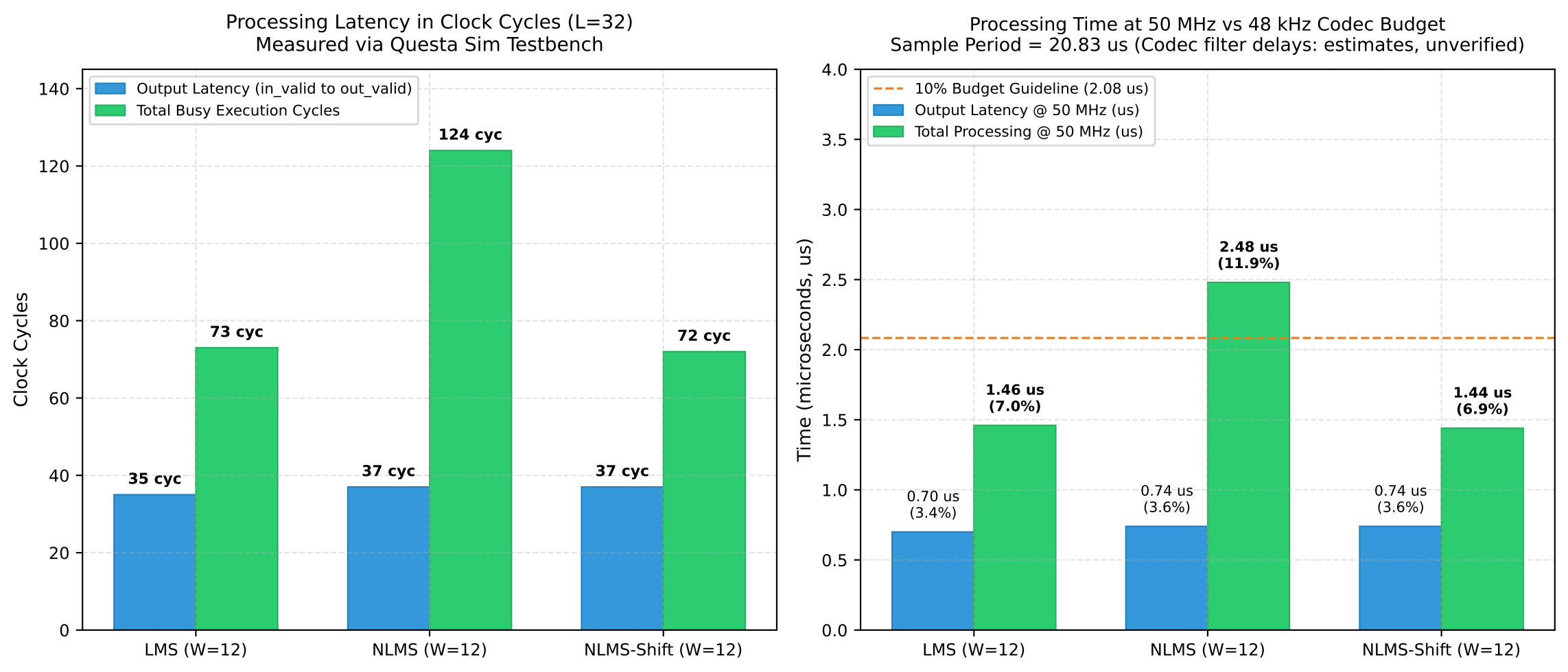
Processing Latency in Clock Cycles (L=32)
Measured via Questa Sim Testbench
Output Latency (in_valid to out_valid)
Total Busy Execution Cycles
Clock Cycles
0
20
40
60
80
100
120
140
35 cyc
73 cyc
37 cyc
124 cyc
37 cyc
72 cyc
LMS (W=12)
NLMS (W=12)
NLMS-Shift (W=12)
Processing Time at 50 MHz vs 48 kHz Codec Budget
Sample Period = 20.83 us (Codec filter delays: estimates, unverified)
10% Budget Guideline (2.08 us)
Output Latency @ 50 MHz (us)
Total Processing @ 50 MHz (us)
Time (microseconds, us)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0.70 us
(3.4%)
1.46 us
(7.0%)
0.74 us
(3.6%)
2.48 us
(11.9%)
0.74 us
(3.6%)
1.44 us
(6.9%)
LMS (W=12)
NLMS (W=12)
NLMS-Shift (W=12)